

PROBLEM STATEMENT REVIEW · 20 AUGUST

Tree Diameter Computation & Visualization System

Finding the longest path between any two nodes of a tree using dynamic programming.

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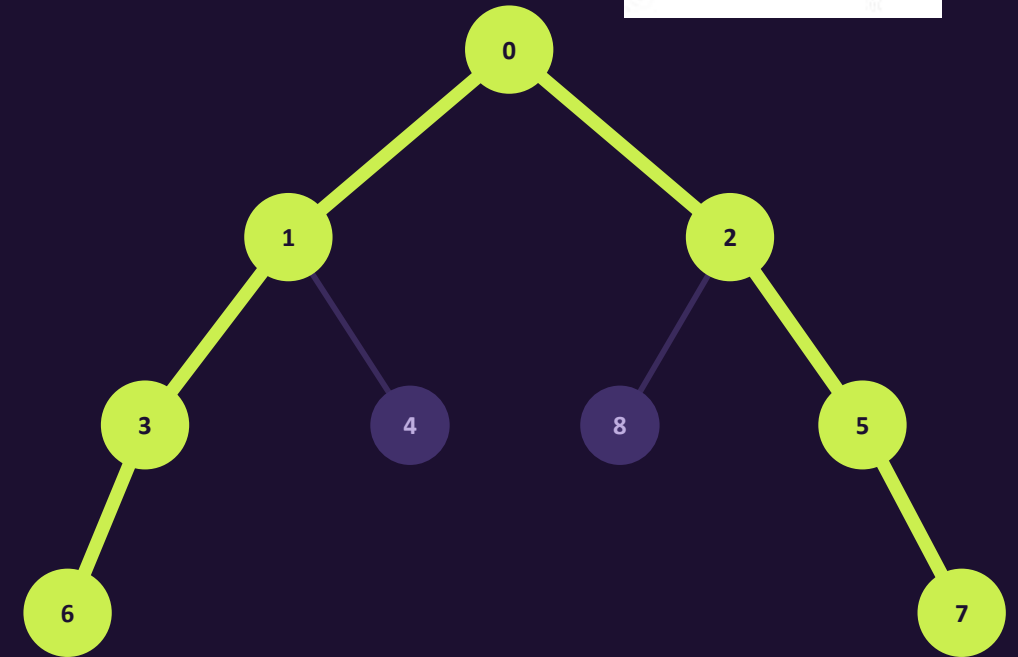
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Highlighted edges mark the diameter path

Abstract

The Tree Diameter Computation and Visualization System is an application designed to compute the diameter of a tree using dynamic programming techniques. The objective is to determine the longest path between any two nodes in a given tree efficiently. The system takes tree edges as input, constructs the corresponding tree structure, and processes it using a tree-based dynamic programming approach. By calculating and reusing the longest paths associated with each node, the system reduces redundant computations and efficiently determines the tree diameter.

The system produces the diameter length, representing the number of edges in the longest path between two nodes. Along with the numerical result, it visualizes the tree and highlights the diameter path, making the result easier to understand. The application is designed to handle trees with up to 1000 nodes, allowing it to process moderately large tree structures efficiently. Overall, the project demonstrates the practical use of dynamic programming, tree traversal, and visualization to solve the tree diameter problem in an efficient and intuitive manner.

1000

nodes supported

Handles moderately large tree structures.

1

traversal pass

Reuses per-node longest paths, no repeated work.

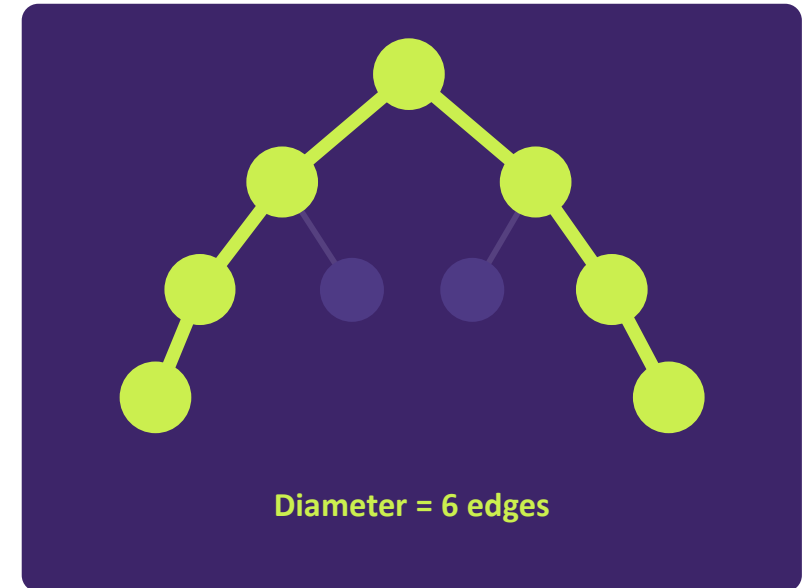
2

outputs

Diameter length plus a highlighted visual path.

Problem Statement

Develop an application to compute the diameter of a tree using dynamic programming techniques. The system should accept the tree edges as input, construct the tree, and determine the longest path between any two nodes efficiently. The calculated diameter length should be displayed as the output. The application should also visualize the tree and highlight the nodes and edges that form the diameter path. The system should be capable of processing trees containing up to 1000 nodes and demonstrate the practical use of dynamic programming and tree traversal techniques.



HOW THE SYSTEM WORKS

1 Input

Read the edge list of the tree

2 Construct

Build the adjacency structure

3 Compute

DP traversal for longest paths

4 Visualize

Show diameter & highlight path

Proposed Functionalities



01

Tree Input

Allow users to enter the nodes and edges of the tree.

02

Tree Visualization

Construct and display the tree based on the given input.

03

Diameter Computation

Use a dynamic programming approach to find the longest path between any two nodes.

04

Path Highlighting

Highlight the nodes and edges that form the diameter path.

05

Result Display

Display the diameter length and the corresponding longest path.

06

Input Validation

Validate the tree structure and support trees with up to 1000 nodes.

Tools and Technologies



CATEGORY	TECHNOLOGY	PURPOSE
● Programming Language	Java	Implement tree structure and dynamic programming algorithm
● Frontend	HTML, CSS, JavaScript	Create the user interface and tree visualization
● Visualization	JavaScript / SVG or Canvas	Display the tree and highlight the diameter path
● Development Tool	VS Code	Write and manage the project code
● Version Control	Git & GitHub	Store, manage, and submit the project
● Algorithm	Dynamic Programming	Efficiently compute the tree diameter

Expected Outcome

The expected outcome of this project is to develop an efficient application that can construct and visualize a tree from the given input edges and compute its diameter using a dynamic programming approach. The system will display the diameter length and the corresponding longest path between two nodes. The diameter path will be visually highlighted in the tree, making the result easy to understand. The application is expected to handle tree structures of up to 1000 nodes efficiently while demonstrating the practical use of dynamic programming and tree traversal techniques.



Correct Result

Accurate diameter length and the longest path between two nodes.



Clear Visualization

The tree is drawn and the diameter path is highlighted on screen.



Efficient Scaling

Smooth handling of tree structures containing up to 1000 nodes.



Concept Demonstrated

Practical application of DP and tree traversal techniques.

Thank You

Questions and suggestions are welcome.

Pranitha M. · Angela · Harshika

Guided by Dr. J Sirisha Devi

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